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My interest in Machine Learning began with a curiosity about how data can be transformed into intelligent decisions. As a B.Tech Information Technology student at Jaypee Institute of Information Technology entering my third year, I have actively explored machine learning through hands on projects that combine software engineering, data processing, and practical problem-solving.

One of the most significant projects I have worked on is **HardwareAI**, an AI powered electronics recommendation platform. The project was designed to help users discover suitable electronic products based on their budget, requirements, and intended use cases. I worked on collecting and preprocessing product data, engineering features from product specifications and reviews, training a Random Forest model using Scikit-learn, and integrating Gemini AI to interpret user requirements and generate recommendation summaries. I also deployed the complete application on Render, which gave me exposure to the full machine learning lifecycle, from model development to production deployment.

Another project that strengthened my understanding of data driven systems was a **Real Estate Data Scraper and Database Pipeline**. Using Python and Selenium, I developed an automated scraper capable of extracting property listings from dynamically rendered websites. I designed a MySQL database schema to store and organize the collected information and implemented data validation and deduplication mechanisms to maintain data quality. This project helped me understand the importance of reliable data collection and preprocessing, which are critical foundations for any machine learning system.

Beyond these projects, I have continuously strengthened my technical foundation through coursework, competitive programming, and self learning. I have solved over 250 problems on LeetCode and completed machine learning courses on Kaggle, which introduced me to concepts such as model training, feature engineering, evaluation metrics, and data preprocessing. These experiences have improved my problem solving ability and helped me approach machine learning challenges with a structured mindset.

While I have gained practical exposure to machine learning workflows, I recognize that there are still significant areas where I want to deepen my understanding. I want to learn more about advanced model architectures, recommendation systems, optimization techniques, model interpretability, and large-scale machine learning systems. I am particularly interested in understanding how machine learning models are designed, evaluated, and deployed in real-world environments that serve millions of users.

I believe Amazon ML Summer School would provide the ideal environment to bridge the gap between my current knowledge and industry-level machine learning expertise. The opportunity to learn directly from researchers and practitioners, engage with challenging concepts, and interact with a community of motivated learners would significantly accelerate my growth.

I am excited about machine learning not only because of its technical depth but also because of its ability to solve meaningful real world problems. With my strong willingness to learn, hands-on project experience, and genuine curiosity about AI systems, I am eager to contribute to and benefit from the Amazon ML Summer School experience.